Electrocardiogram Lab Report

Lab 2

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BME 473: BIOINSTRUMENTATION

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1. Background

The heart generates electrical signals that travel through the atria and ventricles during each cardiac cycle. These electrical signals can be detected at the surface of the skin using electrodes and recorded as an electrocardiogram (ECG). The ECG waveform consists of several components, including the P wave, QRS complex, and T wave, which corresponds to different stages of cardiac depolarization and repolarization.

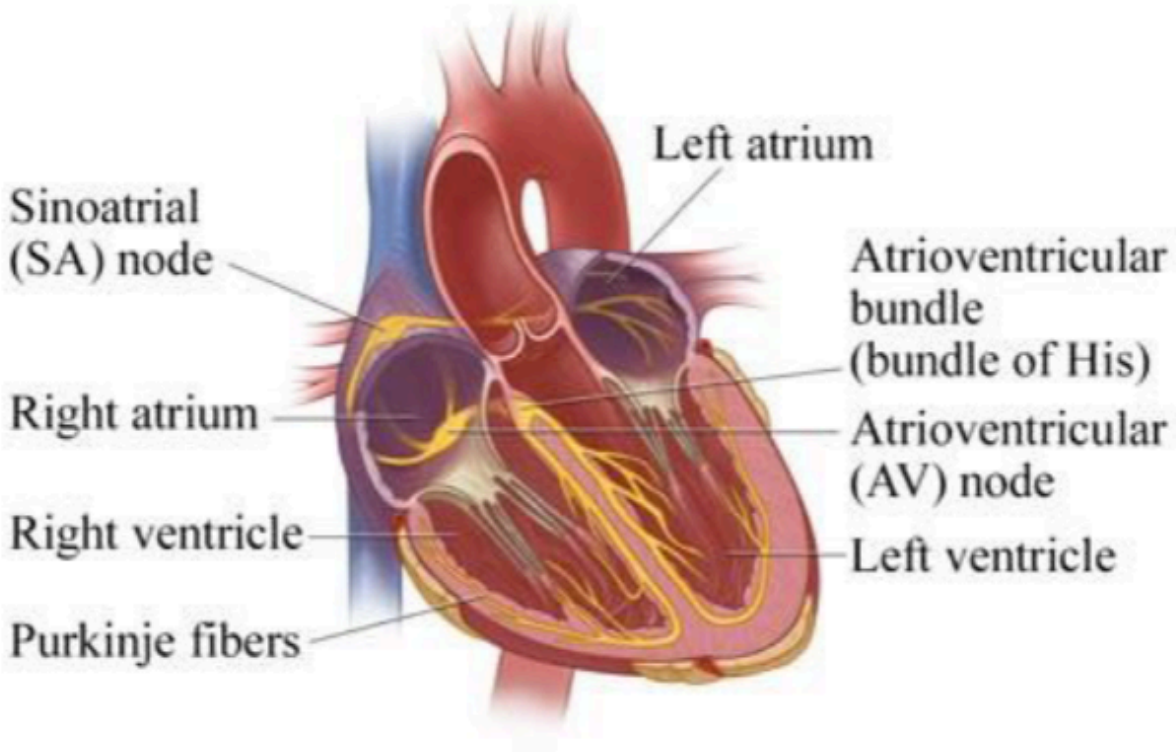


Figure 1: Cross section of the human heart.

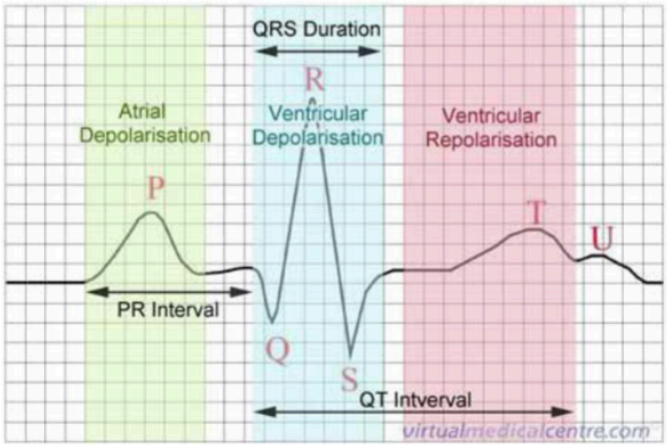


Figure 2: ECG waveform of one heartbeat and primary component.

The P wave represents atrial depolarization, the QRS complex represents ventricular depolarization, and the T wave represents ventricular repolarization. Important ECG parameters such as the RR interval, PR interval, QRS duration, and QT interval can be measured from the waveform and used to evaluate cardiac function.

2. Introduction

The electrocardiogram (ECG) is a noninvasive bioelectrical measurement used to monitor the electrical activity of the heart. Electrical impulses generated by cardiac depolarization and repolarization propagate throughout the body and can be detected at the skin surface using electrodes. The resulting ECG waveform contains characteristic features, including the P wave, QRS complex, and T wave, which provide information regarding cardiac timing and electrical conduction. These features are commonly used to evaluate heart rate, rhythm, and overall cardiac function.

Accurate acquisition of ECG signals requires appropriate signal amplification, filtering, and sampling. Since physiological signals are analog in nature, they must be converted into digital signals before analysis. The sampling frequency must satisfy the Nyquist criterion to ensure that important waveform features are preserved and that aliasing does not occur. In addition, ECG measurements are sensitive to motion artifacts, muscle activity, and variations in electrode-skin contact, all of which can degrade signal quality and interfere with accurate interpretation.

The objectives of this experiment were to investigate the effect of sampling frequency on ECG waveform quality, characterize and quantify ECG waveform parameters, and evaluate the effects of electrode-skin interface disturbances and motion artifacts on ECG recordings. ECG signals were recorded under multiple experimental conditions, including resting, electrode manipulation, and walking. Time intervals and waveform amplitudes were measured and statistically compared with published reference values using onesample twotailed t tests. The results were then used to assess ECG signal quality, determine an appropriate sampling frequency, and evaluate the impact of motion on ECG measurements.

3. Methodology:

3.1 Experimental Setup

This experiment was conducted using an ETH-256 bioinstrumentation amplifier with a C-ISO-256 isolation headstage, a National Instruments CompactDAQ (cDAQ-9174) data acquisition system equipped with an NI-9215 analog input module, ECG surface electrodes, ECG lead wires, and LabVIEW software. The system was used to acquire, amplify, digitize, display, and analyze electrocardiogram (ECG) signals obtained from human subjects.



Figure 3: NI CompactDAQ system, including a USB chassis cDAQ-9174 (left), and an analog input module NI 9215 (right)

The NI CompactDAQ(**Figure 3**) system was used to acquire and digitize the ECG signals for analysis in LabVIEW and the NI 9215 module converted the analog ECG signal into digital data that could be displayed and recorded on a computer.



Figure 4: The ETH-256 Front Panel (top) and Rear Panel (bottom)

The ETH-256 amplifier(**Figure 4**) was used to amplify and filter the low amplitude ECG signal before it was acquired by the CompactDAQ system. The amplifier contains multiple gain settings as well as adjustable high-pass and low-pass filters that allow ECG signals to be recorded with reduced noise. Since ECG signals typically have amplitudes in the microvolt to millivolt range, amplification is required before they can be effectively displayed and analyzed.

The C-ISO-256 isolation headstage(**Figure 5**) was used to ensure safe acquisition of human biopotential signals. The headstage contains three lead connections consisting of a positive lead, a negative lead, and a ground lead. The ECG electrodes were connected directly to the headstage, which provided patient isolation and an internal gain of ×400 before further amplification by the ETH-256 amplifier.



Figure 5: ETH-256C, containing the ETH-256 amplifier and the C-ISO-256 isolated headstage for recording ECG, EMG, and EEG signals from Human

3.2 LabVIEW Data Acquisition Configuration

The ECG signal was acquired through the NI-9215 analog input module and displayed in LabVIEW for real-time monitoring and analysis. Continuous data acquisition was used throughout the experiment to record ECG signals under different conditions.

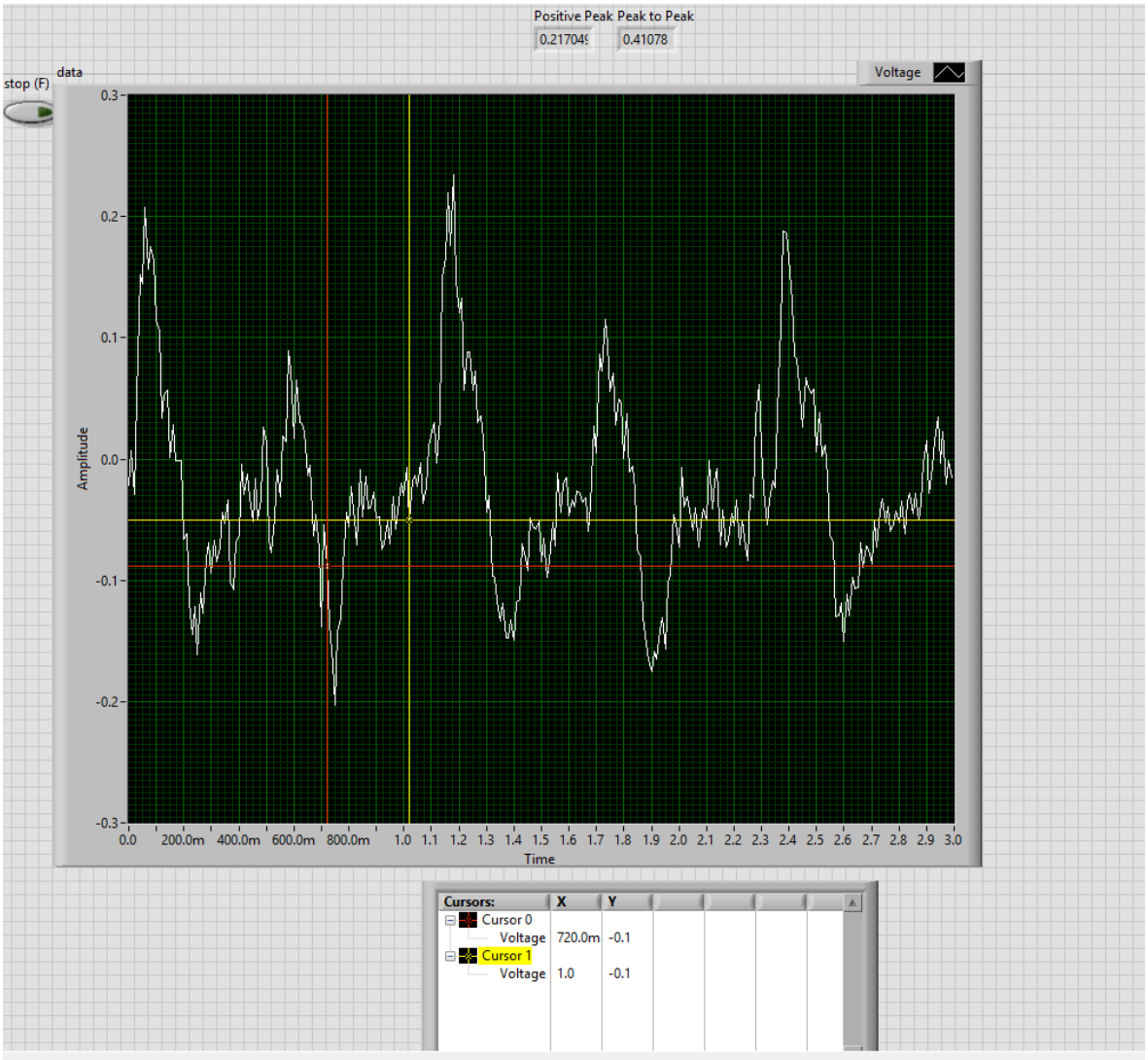


Figure 6: LabVIEW Front Panel used for real time ECG acquisition and waveform visualization.

The LabVIEW front panel(**Figure 6**) displayed the ECG waveform as a voltage(y axis) vs time (x axis) graph. The cursor tools were used to measure ECG amplitudes and time intervals directly from the recorded waveform.

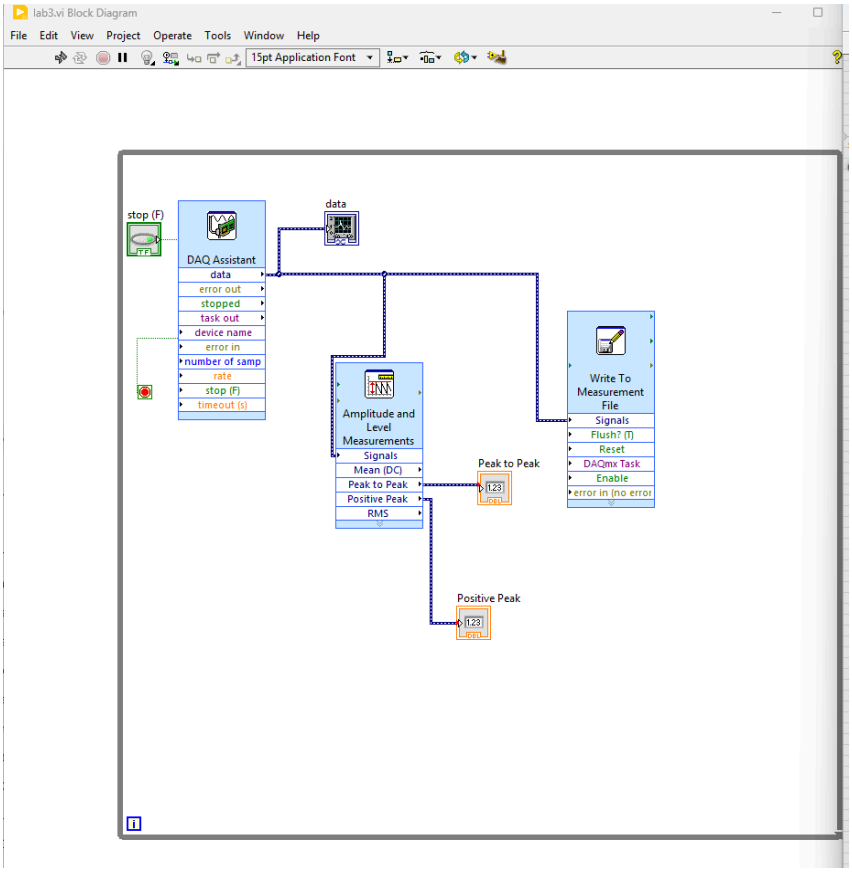


Figure 7: LabVIEW Block Diagram used for ECG signal acquisition and measurement.

The LabVIEW block diagram(**Figure 7**) used the DAQ Assistant Express VI to acquire ECG data from the NI-9215 module. The acquired signal was processed using an Amplitude and Level Measurements Express VI to determine waveform characteristics such as positive peak voltage and peak-to-peak voltage. The ECG data were simultaneously displayed on the front panel and saved to a measurement file for later analysis.

3.3 ECG Electrode Configuration and Data Acquisition

ECG signals were acquired using a Lead II configuration. In this configuration, the negative electrode was connected to the right arm, the positive electrode was connected to the left leg, and the ground electrode was connected to the right leg. The three bipolar limb leads, Leads I, II, and III, form an arrangement known as Einthoven's triangle, shown in Figure 8.

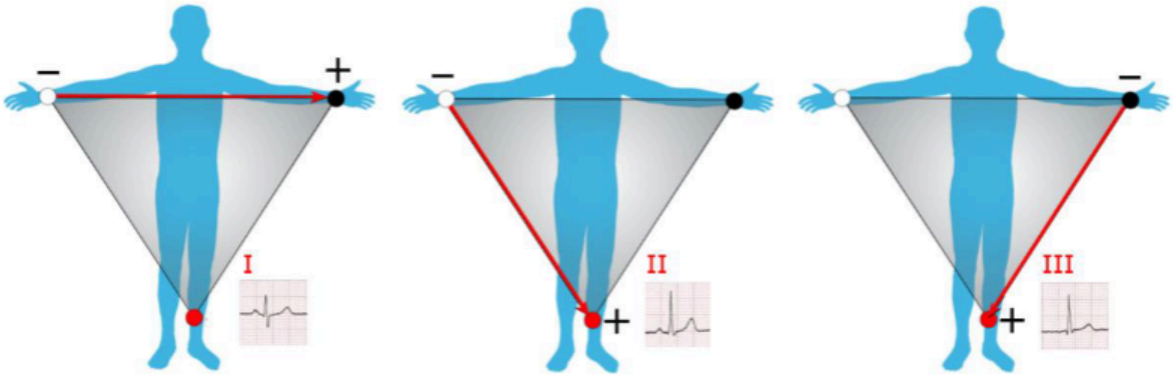


Figure 8: The three bipolar limb leads I, II, and III forming Einthoven's triangle.

The electrodes were connected to the C-ISO-256 isolation headstage and ETH-256 amplifier. The amplified ECG signal was then acquired through the NI-9215 module and displayed in LabVIEW. ECG recordings were collected under resting conditions, electrode press-and-release, pectoral muscle flexion, walking for subsequent analysis and different sampling frequencie.

3.4 Data Analysis

The ECG waveform consists of several components, including the P wave, QRS complex, and T wave, as shown in Figure 9. ECG measurements were obtained directly from the recorded waveform using the cursor tools available on the LabVIEW front panel. The RR interval, PR interval, QRS duration, QT interval, P-wave amplitude, QRS-wave amplitude, and T-wave amplitude were measured for each subject.

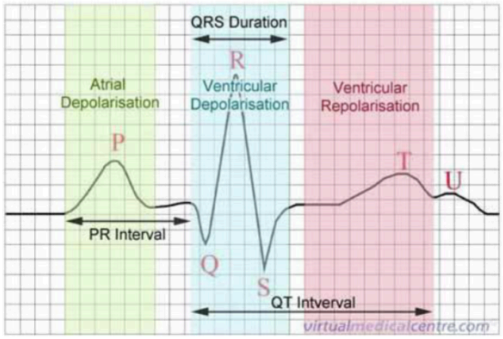


Figure 9: ECG waveform of one heartbeat and primary component

The measurements collected from all subjects were then compiled then analyzed using Microsoft Excel. For each ECG parameter, the mean, standard deviation, minimum value, and maximum value were calculated. The measured values were then compared with published ECG reference values.

A one-sample, two-tailed t-test was then performed to determine whether the class mean differed significantly from the corresponding standard literature value. Since a builtin one-sample t-test function was not available in Excel, the t test was calculated manually using:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x}$ is the sample mean, μ_0 is the literature value, S is the sample standard deviation, and n is the sample size.

For each ECG parameter, the following hypotheses were tested:

$$H_0: \mu = \mu_0$$

$$H_0: \mu \neq \mu_0$$

Where μ represents the class mean and μ_0 epresents the corresponding literature value.

Statistical significance was determined using a significance level of $\alpha = 0.05$

4. Results:

4.1 Sampling and the Nyquist Theorem

Sampling Rate(Hz)	ECG visible
1000	Yes
750	Yes
500	Yes
200	Yes
50	No
10	No

Table 1: Effect of Sampling Rate on ECG Visibility

As shown in Table 1, the ECG signal was clearly visible at sampling rates of 1000 Hz, 750 Hz, 500 Hz, and 200 Hz. At 50 Hz and 10 Hz, the signal quality decreased and important ECG features could not be identified. Based on these results, 200 Hz was the minimum acceptable sampling rate, while 500-1000 Hz provided the clearest ECG recordings.

4.2 PQRST Waveform Characterization and Statistical Analysis

ECG interval and amplitude measurements were collected from 20 subjects. The RR interval, PR interval, QRS duration, QT interval, P-wave amplitude, QRS-wave amplitude, and T-wave amplitude were measured from each subject's ECG recording. The raw measurements are shown in Table 2.

Subject	Gender	RR (ms)	PR (ms)	QRS (ms)	QT (ms)	P Wave (mV)	QRS Wave (mV)	T Wave (mV)
AC	F	669.91	223.30	87.38	359.23	0.1529	0.9091	0.2934
AN	F	757.29	150.48	67.97	373.79	0.0682	0.2888	0.1736
FS	F	596.09	122.70	75.13	315.48	0.0675	0.5020	0.1673
MS	F	682.42	168.39	84.19	367.79	0.0318	0.4879	0.1646
LG	F	889.12	153.74	66.43	282.90	0.0309	0.6258	0.1756
HH	F	856.23	166.13	127.80	466.46	-	0.4283	0.3695
LL	F	667.13	255.95	67.13	348.25	0.0600	0.3114	0.1096
AG	M	1147.92	150.00	80.00	335.41	0.0950	0.2325	0.2813
CD	M	735.00	125.00	141.00	380.00	0.1488	0.5538	0.2967
JC	M	844.20	203.00	58.40	371.70	0.4450	0.2250	0.1130
SK	M	1200.00	100.00	100.00	400.00	0.0750	0.6750	0.1500
DS	M	671.00	167.90	68.10	312.90	0.1000	0.4000	0.2000
TJ	F	849.57	111.77	84.95	362.15	0.0424	0.4065	0.1571
BT	M	723.00	141.00	56.00	243.00	0.1665	1.9900	0.2500

TD	F	967.20	100.91	65.57	383.30	0.0997	0.6657	0.1856
SSK	M	830.00	231.00	80.00	423.00	0.1800	1.2000	0.3500
JS	M	862.62	153.35	223.64	421.73	0.1335	1.1501	0.5335
NK	F	900.40	165.15	90.43	326.34	0.2570	1.9346	0.2310
JL	F	824.50	134.20	62.37	267.91	0.0980	1.0080	0.1568
GM	F	755.00	225.00	150.00	380.00	0.1000	0.3000	0.1000
JS	M	813.00	127.00	63.00	291.00	0.0575	1.7500	0.4730

Table 2: Raw ECG Measurements Collected from All Subjects

The duration measurements were analyzed across all subjects and compared with published reference values using a one-sample, two-tailed t-test. The results are summarized in Table 3.

Parameter	**Mean ± SD**	**Min**	**Max**	**Literature Value**	**t Statistic**	**p Value**	**Significant?**
RR Interval (ms)	821.03 ± 150.25	596.09	1200.00	850	-0.862	0.399	No
PR Interval (ms)	160.76 ± 44.12	100.00	255.95	180	-1.950	0.066	No
QRS Duration (ms)	90.45 ± 40.22	56.00	223.64	80	1.162	0.260	No
QT Interval (ms)	352.97 ± 54.94	243.00	466.46	400	-3.829	0.001	Yes

Table 3: Statistical Analysis of ECG Duration Measurements

The duration measurements were analyzed across all subjects and compared with published reference values using a one-sample, two-tailed t-test. The results are summarized in Table 3.

Parameter	Mean ± SD	Min	Max	Literature Value	t Statistic	p Value	Significant?
P Wave (mV)	0.120 ± 0.095	0.0309	0.445	0.10	0.965	0.347	No
QRS Wave (mV)	0.764 ± 0.552	0.225	1.99	1.08	-2.562	0.019	Yes
T Wave (mV)	0.235 ± 0.117	0.100	0.534	0.30	-2.483	0.023	Yes

Table 4: Statistical Analysis of ECG Amplitude Measurements

ECG interval and amplitude measurements were collected from 20 subjects and compared with published reference values using a one-sample, two-tailed t-test. As shown in Table 3, the RR interval, PR interval, and QRS duration were not significantly different from the literature values ($p > 0.05$). However, the QT interval was significantly different from the reference value ($p = 0.001$). As shown in Table 4, the P-wave amplitude was not significantly different from the literature value ($p = 0.347$), while the QRS-wave amplitude and T-wave amplitude were significantly different from the reference values ($p < 0.05$). These differences may be due to physiological variation among subjects, electrode placement, signal noise, and measurement uncertainty during ECG analysis.

4.3 ECG Signal Quality Under Different Conditions

ECG recordings were collected under resting conditions, hand pressing, electrode press-and-release, and walking. The resulting waveforms were compared to evaluate the effect of motion and electrode-skin interface disturbances on ECG signal quality. Representative ECG recordings for each condition are shown below. For this section thes settings were to:

Settings	Value
Headstage LFP	10k
Amplifier Gain	x10
HPF setting	DC
LPF setting	10k
Sampling Rate(DAQ Assist)	1k
Samples to Read	500

Table 5: Acquisition settings used during motion artifact analysis

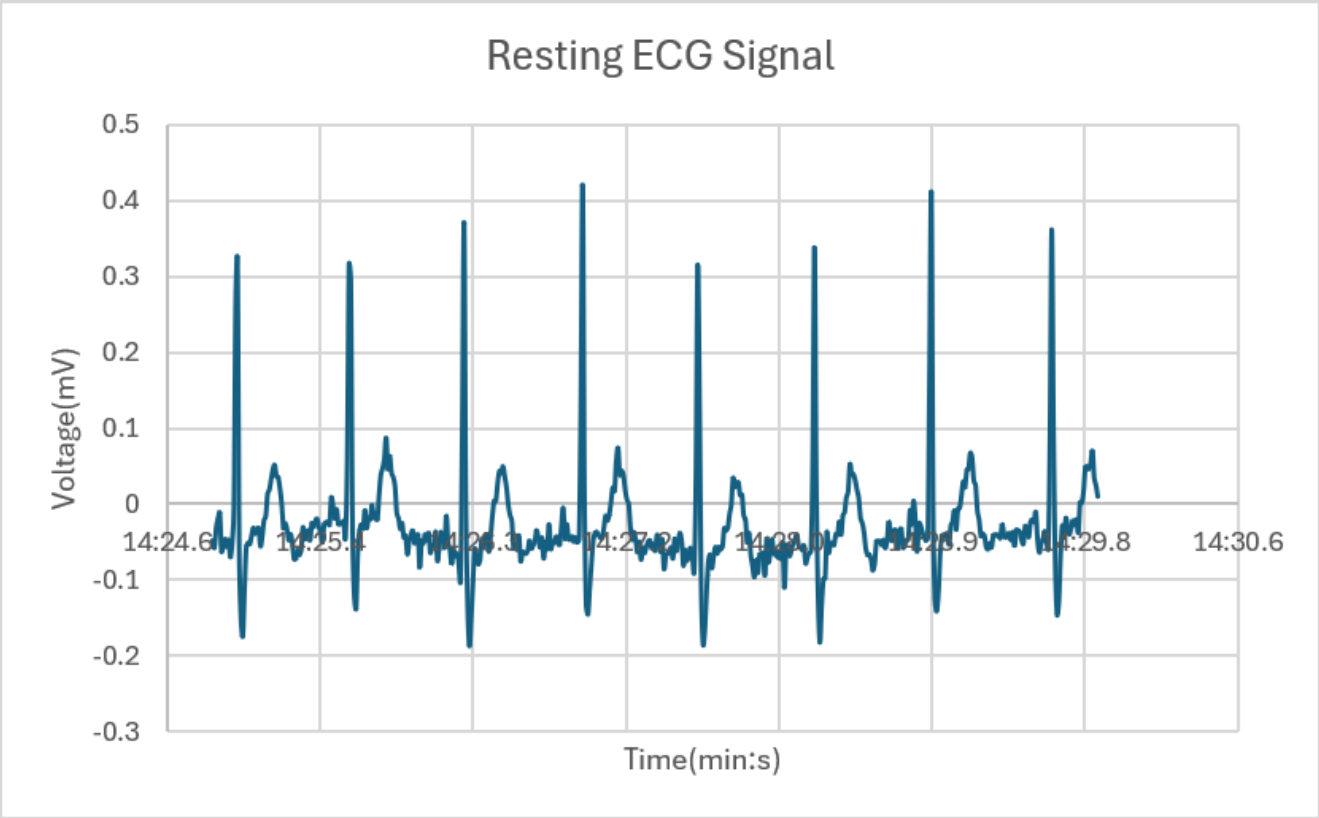


Figure 10: Resting ECG Signal

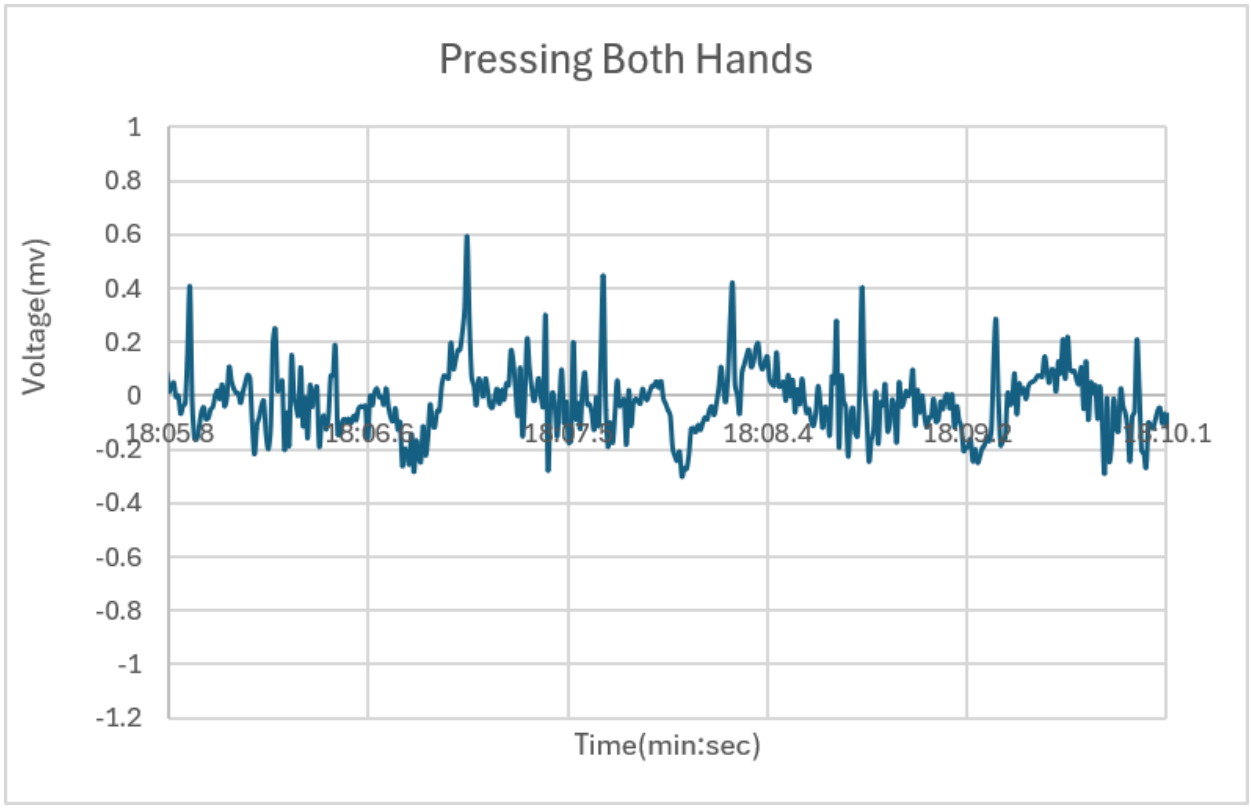


Figure 11: ECG Signal During Hand Pressing

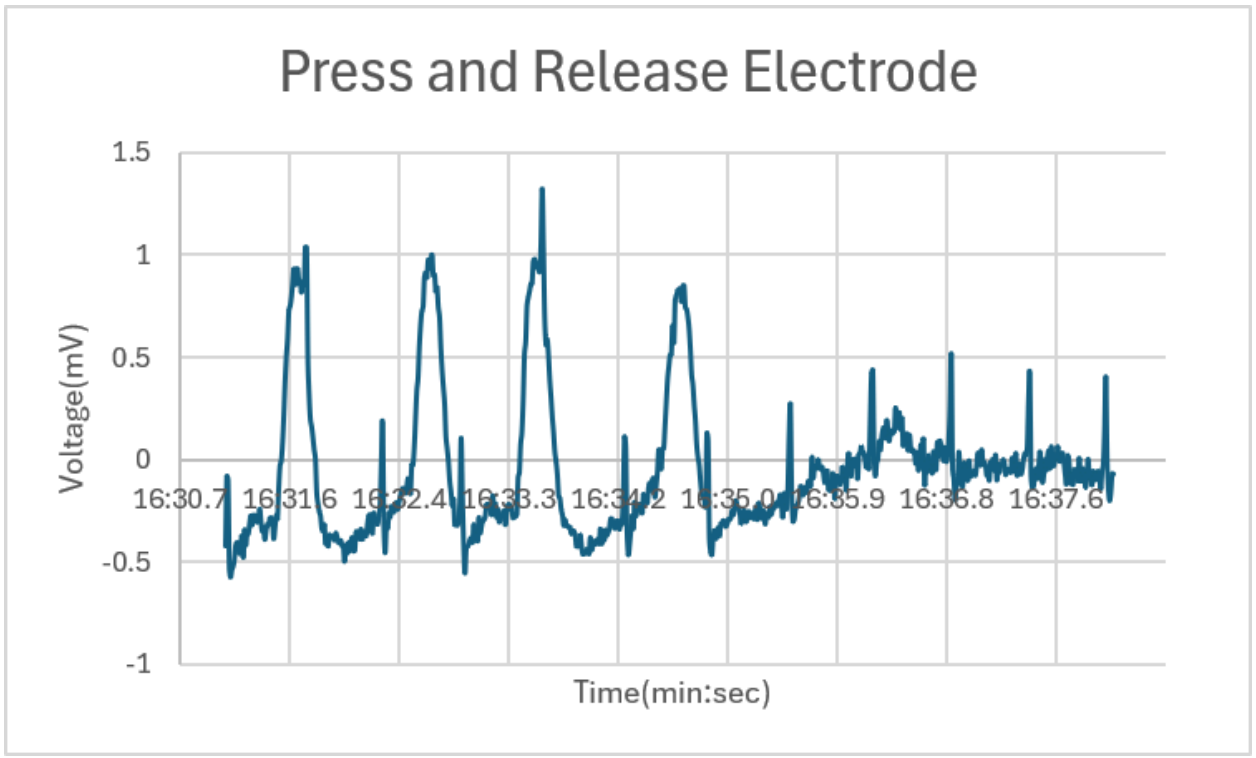


Figure 12: ECG Signal During Electrode Press-and-Release

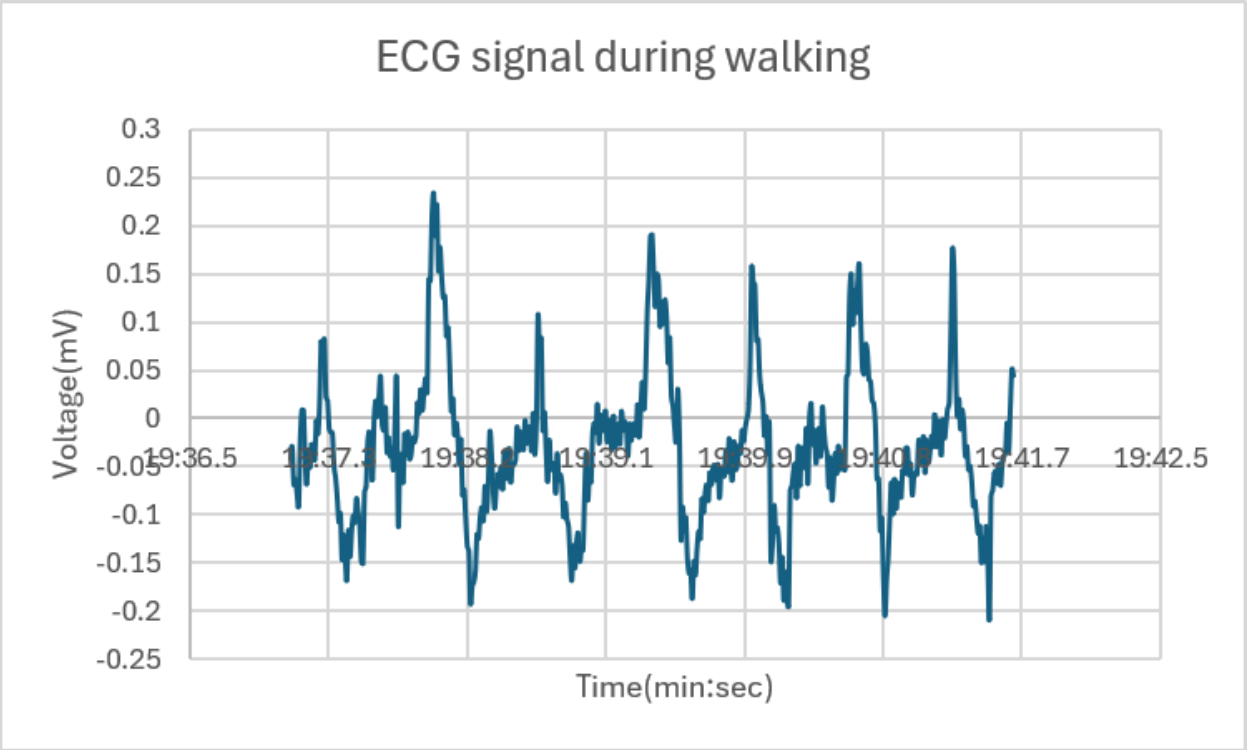


Figure 13: ECG Signal During Walking

The resting ECG produced the clearest waveform and the most stable baseline. Hand pressing introduced additional fluctuations and low-level noise into the recording. The electrode press-and-release condition generated the largest transient voltage spikes due to changes in electrode-skin contact and electrode-skin impedance. Walking introduced baseline drift and motion artifacts caused by body movement and electrode displacement. Overall, ECG signal quality decreased as movement increased, with the press-and-release condition producing the greatest waveform distortion.

5. Conclusion:

This experiment successfully demonstrated the acquisition and analysis of ECG signals using the ETH-256 bioinstrumentation amplifier, C-ISO-256 isolation headstage, NI CompactDAQ system, and LabVIEW software. ECG intervals and waveform amplitudes were measured and compared with published reference values using statistical analysis.

The sampling rate investigation showed that ECG signals could be clearly identified at sampling rates of 1000 Hz, 750 Hz, 500 Hz, and 200 Hz. At 50 Hz and 10 Hz, important ECG features could no longer be clearly distinguished. These results showed that a sufficient sampling rate is necessary to accurately record ECG signals and satisfy the Nyquist sampling criterion.

Statistical analysis showed that the RR interval, PR interval, QRS duration, and P-wave amplitude were not significantly different from the corresponding literature values ($p > 0.05$). However, significant differences were observed for the QT interval, QRS-wave amplitude, and T-wave amplitude ($p < 0.05$). These differences may be due to normal physiological variation among subjects, electrode placement, signal noise, and measurement uncertainty.

The ECG recordings collected under different conditions demonstrated the effect of motion and electrode-skin interface disturbances on signal quality. The resting ECG produced the clearest waveform and the most stable baseline. Hand pressing introduced additional fluctuations in the signal, while electrode press-and-release generated the largest voltage spikes due to changes in electrode contact. Walking introduced motion artifacts and baseline drift caused by body movement. Overall, ECG signal quality decreased as movement increased, highlighting the importance of proper electrode placement and minimizing motion during ECG acquisition.

This experiment demonstrated how ECG signals can be measured, analyzed, and affected by sampling rate and motion artifacts. Understanding these factors is important for obtaining accurate and reliable ECG recordings.